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Command Prompt
Microsoft Windows [Version 10.0.19043.1151]
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C:\Users\ABHIRAM>pip3 install setuptools
Requirement already satisfied: setuptools in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (49.2.1)

C:\Users\ABHIRAM>pip3 install -U git+https://github.com/GoSecure/malboxes.git#egg=malboxes
Collecting malboxes
  Cloning https://github.com/GoSecure/malboxes.git to c:\users\abhiram\appdata\local\temp\pip-install-oh41sexn\malboxes_e88c6293bcfb4d319144bf1f6f736b28
  Running command git clone -q https://github.com/GoSecure/malboxes.git 'C:\Users\ABHIRAM\AppData\Local\Temp\pip-install-oh41sexn\malboxes_e88c6293bcfb4d319144'
  Resolved https://github.com/GoSecure/malboxes.git to commit 32d358f7007b23018cd898454d62dc2198df3682
Requirement already satisfied: appdirs in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from malboxes) (1.4.4)
Requirement already satisfied: Jinja2 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from malboxes) (3.0.1)
Requirement already satisfied: jsmn in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from malboxes) (2.2.2)
Requirement already satisfied: boto3 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from malboxes) (1.18.14)
Requirement already satisfied: botocore<1.22.0,>=1.21.14 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from boto3->malboxes) (1.21.14)
Requirement already satisfied: jmespath<1.0.0,>=0.7.1 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from boto3->malboxes) (0.10.0)
Requirement already satisfied: s3transfer<0.6.0,>=0.5.0 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from boto3->malboxes) (0.5.0)
Requirement already satisfied: urllib3<1.27,>=1.25.4 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from botocore<1.22.0,>=1.21.14) (1.25.4)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from botocore<1.22.0,>=1.21.14) (2.8.1)
Requirement already satisfied: six>=1.5 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from python-dateutil<3.0.0,>=2.1->botocore) (1.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\abhiram\appdata\local\programs\python\python39\lib\site-packages (from Jinja2->malboxes) (2.0.1)

C:\Users\ABHIRAM>

```

Figure 3: Installation of Malboxes and setuptools through CLI

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C:\Users\ABHIRAM>malboxes build <template>

```

Figure 4: Command for checking Malboxes creation

configuration

- PowerShell recipes to disable security features (Defender, auto-updates, firewall, etc)
- A set of malware analysis focused tools installed using the Windows package manager Chocolatey
- A Vagrant file generator so that the VMs can be managed with Vagrant

## Tools that come with the Malboxes VM

The following tools are installed as part of the build process.

- **WinDBG:** A well-known name for the Windows debugger, it offers regular updates, functionality, and online support. It has a graphical user interface (GUI) that displays the debugger output, as well as stacks and registers. It is used to examine memory dumps in both kernel and user mode.
- **Complete Sysinternal suite:** This suite of tools consists of a collection of Windows apps that can be downloaded for free from Microsoft Technet's Sysinternals area. They are all portable, which means that

they can be placed on a Flash drive and used from any PC, eliminating the need of installing them.

- **Fiddler:** A debugging proxy server tool that allows logging, analysing, and changing HTTP and HTTPS traffic between your computer and a Web server/s.
- **Wireshark with Npcap (a Windows 10 compatible winpcap replacement):** Npcap is a Windows version of the libpcap library that includes a packet capture driver. This library can be used by Wireshark on Windows to record live network data.
- **IDA Pro remote debuggers (if IDA Pro is installed on the host):** The 'debugger server' is the computer that runs the debugging application. To debug viruses, Trojans, and malware, the debugger client is kept as far away from the compromised computer as feasible. The following changes are made to the guest operating system:
  - Automatic updates disabled
  - Windows Defender disabled

## Why Malboxes?

It is important to analyse malware in a safe environment. Malboxes, with its tools, helps us do just that.

- Malboxes builds everything required in the analysis toolkit, in

the following way:

1. Creates and allocates a safe environment for analysis
  2. Sandboxes the system from the production environment
  3. Installs behaviour and code analysis tools
- Malboxes helps to overcome faulty malware analysis consequences, which can range from the leakage of internal information, products, or licences to the identification of an organisation's IP addresses and erasure from the infected computer.
  - Because Malboxes includes Chocolatey, a single command can be used to install anything from the large Chocolatey software package gallery. In the configuration file of Malboxes, you can change the default list of packages installed.
  - Malboxes offers the advantage of coming with tools preloaded and may be deployed with the help of a few commands, avoiding hours of effort in installing these individually.
  - Malboxes is built around Python 3 and supports the Windows, Mac and Linux operating systems.

## Installation steps on a Windows machine

Requirements

- Python 3.3+
- Packer